

retry error:

```
import tenacity
from tenacity import *
@retry(stop=stop_after_attempt(5))
def retry_with_attempts():
    print("Stopping after 5 attempts")
    raise Exception("retry with attempts")
```

Retry with time duration: The retry loop runs for 20 secs in the following code:

```
import tenacity
from tenacity import *
@retry(stop=stop_after_delay(20))
def retry_stop_after_20_s():
    print("Stopping after 20 seconds")
    raise Exception("retry until 20 sec is completed")
```

Waiting before retries

Many times, we may need to wait before the retry is attempted. This feature is specifically useful in areas where throttling of requests needs to be handled.

```
import tenacity
from tenacity import *
```

```
@retry(wait=wait_fixed(2), stop=(stop_after_delay(12) | stop_
after_attempt(6)))
def retry_wait_between_5_s():
    print("Wait 2 seconds between retries")
    raise Exception("retry")
```

In the above example, we have combined wait and attempt levels along with the duration of retry. In this case, the code will try to retry six times, post which a retry error will be raised.

There are different strategies of waits that are adopted like *wait_random*, *wait_exponential*, *wait_chain*, etc. You can explore these and decide the best choice for your application code.

Condition for retry

Many times, we would like the retry to happen only for a certain condition or error message. For example, we may want to retry a function only if we have an *IOError* raised.

```
import tenacity
from tenacity import *
import random
@retry(retry=retry_if_exception_type(IOError))
def retry_io_error():
    print("Retry forever with no wait if an IOError occurs,
```

```
/usr/local/bin/python3 /Applications/PyCharm.app/Contents/plugins/python/helpers/pydev/pydevd.py --multiprocess
--qt-support=auto --client 127.0.0.1 --port 53357 --file /Users/Srikanth_Mohan/Learning/learn_tenacity.py
Connected to pydev debugger (build 222.3345.131)
Retry forever with no wait if an IOError occurs, raise any other errors
Retry forever with no wait if an IOError occurs, raise any other errors
Retry forever with no wait if an IOError occurs, raise any other errors
Retry forever with no wait if an IOError occurs, raise any other errors
python-BaseException
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/concurrent/futures/_base.py", line 438, in
    result
    return self.__get_result()
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/concurrent/futures/_base.py", line 398, in
    __get_result
    raise self._exception
  File "/Library/Frameworks/Python.framework/Versions/3.10/lib/python3.10/site-packages/tenacity/_init_.py", line
    407, in __call__
    result = fn(*args, **kwargs)
  File "/Users/Srikanth_Mohan/Learning/learn_tenacity.py", line 9, in retry_io_error
    raise random.choice([IOError, Exception])
Exception

Process finished with exit code 1
```

Figure 1: Output of retry with IOError